

Project: 50 Python One-Liners Everyone Should Know

Problem Statement

Python provides many powerful built-in functions, expressions, and library features that allow complex tasks to be completed in just one line of code. This project demonstrates 50 practical Python one-liners that improve productivity, readability, and programming efficiency. Students will explore these examples, understand how they work, and practice applying them to real-world programming tasks.

Objective

Create a Python notebook containing 50 useful Python one-liners. Each one-liner should solve a specific problem or perform a common task using concise and efficient Python syntax.

Requirements

- Write and execute 50 Python one-liners.
- Include a short comment describing each one-liner.
- Cover different Python concepts such as strings, numbers, lists, dictionaries, sets, sorting, file handling, lambda functions, comprehensions, and standard libraries.
- Display the output for every one-liner.
- Organize the notebook with proper numbering and formatting.

Concepts Covered

- String Manipulation
- List Comprehensions
- Lambda Functions
- Dictionary Operations
- Set Operations
- Sorting
- Mathematical Operations
- File Handling
- Collections Module
- Built-in Functions
- Python Standard Library
- Data Processing

Expected Learning Outcomes

- Understand concise Python syntax.
- Improve coding efficiency.
- Learn useful built-in functions and modules.
- Apply one-liners to solve common programming tasks.

- Develop clean and readable Python code.